CprE 458/558: Real-Time Systems

Term Project List, Requirements

Project Requirements and Schedule

- Project Plan
  - 2 page report
    - Title, team members, type of project (simulation or implementation)
    - problem statement, identify the relevant algorithms/protocols, expected output/results of the project, list of references

- Final Project Report + Project Presentation slides
  - Due: Please refer to Canvas (typically at the end of Prep Week)
  - Submitted via Canvas

The Requirements for Project Plan, Final Report, and Presentation will be posted in Canvas as part of the assignments.

Project Types (Categories of projects)

- The projects can be in one of following forms:
  - **Type 1: GUI simulator** -- similar to App-like tool where input can be provided, output is visually shown
  - **Type 2: Implementation** -- using real hardware/software platform
  - **Type 3: Simulation** (performance) studies – using an existing simulator (or develop a simulator) and conducting systematic performance evaluation under different workloads. This typically involves independent reading of research paper(s) or industry report(s)
- CprE 458 team can choose Type 1, 2, or 3 (i.e., any type)
- CprE 558 team can choose only Type 2 or 3.
- Team size: 1-member team or 2-member team
- The project scope should be proportional to the team size

Sample project list, with suggested "Type"

- **Resource Access Control**
  - Type 1 or Type 2: RMS with Priority inheritance, Priority ceiling or priority ceiling emulation protocols.
  - Team size: 1 (inheritance); 2 (priority inheritance and Priority ceiling)
  - Show priority inversion, deadlock, deadlock-avoidance scenarios

- Feedback-based EDF:
  - Type 2 or Type 3: Feedback based EDF scheduler.
  - Team size: 2 (EDF and Feedback EDF)
  - Show performance under different overload scenarios
- Imprecise computations – (m,k)-firm model
  - Type 2 or Type 3: Overload handling using (m,k)-firm model
  - Team size: 1 (RMS and m,k_RMS), 2 (RMS, m,k_RMS and detailed analysis)
  - Show performance under different overload scenarios

- Energy-aware real-time Scheduling
  - Type 1, Type 3: DVS techniques -- inter-task DVS, or intra-task DVS
  - Team size: 2 (Inter-task DVS – 3 algorithms); Team size: 1 (intra task DVS)

- Energy-aware scheduling in sensor networks
  - Type 2, Type 3: Modulation scaling or network coding, or other techniques
  - Team size 1 or 2 (network coding), Team size 1 (modulation scalring)

Project list (contd.)

- Real-Time Wide-Area NetworkS (WAN)
  - Type 2, 3: QoS routing algorithms
  - Team Size: 1 or 2 (2 or 3 algorithms and evaluation)
- Real-Time Wide-Area NetworkS (WAN)
  - Type 2, 3: Packet scheduling algorithms
  - Team Size: 1 or 2 (2 or 3 algorithms and evaluation

- Multiprocessor scheduling
  - Type 3: Multiprocessor scheduling
  - Team size: 2 (Myopic scheduling or any algorithm)
- IoT Applications
  - Type 2, Type 3: IoT applications or IoT protocols
  - Team size: 1 or 2 (depends on the application and protocol)
- Fault-tolerance in safety-critical systems
  - Type 2, Type 3: Fault-Tolerant design Techniques
  - Team size: 1 or 2 (depends on the application)
- Real-Time Local-Area Networks (LAN)
  - Type 1, 2: MAC protocols
  - Team Size: 2 (2 protocols and evaluation)

Project List (contd.)

- Other Projects: Your own choice of project focusing on real-time systems, networks, or applications.

Tools/Platforms to consider:

- Free RTOS or VxWorks or any other "real-time" OS – relevant to single processor scheduling, resource access control, overload handling projects
- Android OS – Power management project (Schedulers and Governors)
- Sensor network projects – Raspberry PI or other platforms (e.g., Berkley Motes)
- CAN Bus simulator or platform – CAN bus protocol study
- Embedded platforms in CprE 288/388/488 classes – e.g., iRobot's Roomba platform used in CprE 288.
- Real-Time IoT platforms (with cloud connectivity)